2.2 Inverse of a matrix

- An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C such that $CA = I$ and $AC = I$ where $I = I_n$.
- In this case, C is an inverse of A.
- A matrix that is not invertible sometimes called a singular matrix, and an invertible matrix is called a nonsingular matrix.

Ex:

If $A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$ and $C = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$

A. $CA = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}\begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2\cdot -7 + 5\cdot 3 & 2\cdot -5 + 5\cdot 2 \\ -3\cdot -7 + -7\cdot 3 & -3\cdot -5 + -7\cdot 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

C. $AC = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = \begin{bmatrix} -7\cdot 2 + -5\cdot -3 & -7\cdot 5 + -5\cdot -7 \\ 3\cdot 2 + 2\cdot -3 & 3\cdot 5 + 2\cdot -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A \cdot C = C \cdot A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$. Thus, $C = A^{-1}$

(inverse of A is denoted by A^{-1})

Theorem: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$ then A is invertible and

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, then A is not invertible.

The quantity $ad - bc$ is called the determinant of A.

Ex: $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$ $A^{-1} = \frac{1}{3\cdot 6 - 5\cdot 4} \begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}}_{A^{-1}}$ $\quad -2$

Theorem: If A is an invertible $n \times n$ matrix, then for each b in $\mathbb{R}^n$, the equation $Ax = b$ has the unique solution $x = A^{-1} \cdot b$.

Remember $AB \neq BA$, in general.

$$Ax = b \Rightarrow \underbrace{A^{-1}A}_{I_n} x = A^{-1}b \Rightarrow x = A^{-1} \cdot b$$

Ex: $3x_1 + 4x_2 = 3$
$5x_1 + 6x_2 = 7$. Use inverse of A to solve the system.

$x = A^{-1}b$. See that $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$ and remember that $A^{-1} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}$

Then

$$x = A^{-1}b = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

Theorem:

a. If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$

b. If A and B are $n \times n$ invertible matrices, then so is AB. The inverse of AB is the product of the inverses of A and B in the reverse order. That is $(AB)^{-1} = B^{-1} \cdot A^{-1}$.

c. $(A^T)^{-1} = (A^{-1})^T$

Theorem:

An $n \times n$ matrix is invertible if and only if A is row equivalent to I_n, and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1}.

$$A \rightarrow I_n$$

$$I_n \rightarrow A^{-1}$$

$A_{n\times n} \to [A\ I]_{n\times 2n}$

Algorithm for finding A^{-1}.

- Row reduce the augmented matrix $[A\ I]$. If A is row equivalent to I, then $[A\ I]$ is row equivalent to $[I\ A^{-1}]$. Otherwise, A does not have an inverse.

Ex: $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ $\quad Ax = 0 \Rightarrow x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$[A\ \ I] = \left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right] \sim \begin{matrix} R_2 \\ R_1 \\ \\ \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \begin{matrix} \\ \\ R_3 - 4R_1 \to \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array}\right] \qquad \begin{matrix} \\ \\ R_3 + 3R_2 \to \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array}\right]$$

$$\begin{matrix} \\ \\ R_3/2 \to \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{array}\right] \qquad \begin{matrix} R_1 - 3R_3 \to \\ R_2 - 2R_3 \to \\ \\ \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{array}\right]$$

I $\qquad$ A^{-1}

$[A\ \ I] \to [I\ \ A^{-1}]$ $\qquad$ $Ax = 0$

Thm, $A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}$

$A^{-1} \cdot A = A \cdot A^{-1} = I_3$ $\qquad$ You can check at home :)

The invertible matrix theorem:

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A, the statements are either all true or all false.

a. A is an invertible matrix.
b. A is row equivalent to the $n \times n$ identity matrix.
c. A has n pivot positions.
d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
e. The columns of A form a linearly independent set.
f. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
g. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
h. The columns of A span $\mathbb{R}^n$.
i. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
j. There is an $n \times n$ matrix C such that $CA = I$.
k. There is an $n \times n$ matrix D such that $AD = I$.
l. A^T is an invertible matrix.

Notes: The invertible matrix theorem divides the set of all $n \times n$ matrices into two disjoint classes: the invertible (nonsingular) matrices and the noninvertible (singular) matrices. Each statement in the theorem describes a property of every $n \times n$ invertible matrix. The negation of a statement in the theorem describes a property of every $n \times n$ singular matrix.

- The invertible matrix theorem applies only to square matrices.